

5G RAN

Access Class Control Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
 - Changes in functions and their corresponding parameters
- Editorial changes
 - Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 01 (2025-03-10).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added the parameter that specifies the measurement mode of the number of RRC_CONNECTED UEs in dynamic access class control. For details, see: • 4.1.3 Dynamic Access Class Control • 4.2.2 Impacts • 4.3.2.2 Mutually Exclusive Functions • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added the NRDUCellUacBar.RrcUserNumCountMode parameter.	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

Feature ID	Feature Name	Chapter/Section
FBFD-051102	Access Control	4 Unified Access Control 5 Access Control Based on Operator-specific Reservation

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Access class control	Supported only in SA networking	4 Unified Access Control 5 Access Control Based on Operator-specific Reservation

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Access class control	Supported in both low and high frequency bands, with the following differences: TDD: Access barring factor adaptation is supported only in high frequency bands, not in low frequency bands.	4 Unified Access Control 5 Access Control Based on Operator-specific Reservation

3 Overview

In section 4.5 "Unified access control" of 3GPP TS 24.501 V16.5.0, 16 access IDs are defined, corresponding to access classes 0 to 15. Access class 0 corresponds to common UEs; access classes 1 to 2 and access classes 11 to 15 correspond to UEs performing special services or special UEs. Each UE may have one or more access IDs.

Section 6.22 "Unified access control" in 3GPP TS 22.261 V15.6.0 defines 64 access classes, numbered from 0 to 63. According to section 4.5 "Unified access control" in 3GPP TS 24.501 V18.0.1, services can be mapped to specific access classes based on mapping rules. Access class control is not required for UEs of access class 0. Therefore, the gNodeB only needs to deliver the control information about access classes 1 to 63.

Access class control includes unified access control (UAC) and access control based on operator-specific reservation.

- UAC: Network congestion occurs when a large number of UEs simultaneously access a network, causing signaling impact on the network. With UAC, the gNodeB delivers access control parameters to control the access of UEs in idle or inactive mode in a cell, thereby relieving network congestion. For details, see [4 Unified Access Control](#).
- Access control based on operator-specific reservation: When operators want to reserve cell resources for special UEs in some special scenarios, a cell can be reserved for specified operators. For details, see [5 Access Control Based on Operator-specific Reservation](#).

4 Unified Access Control

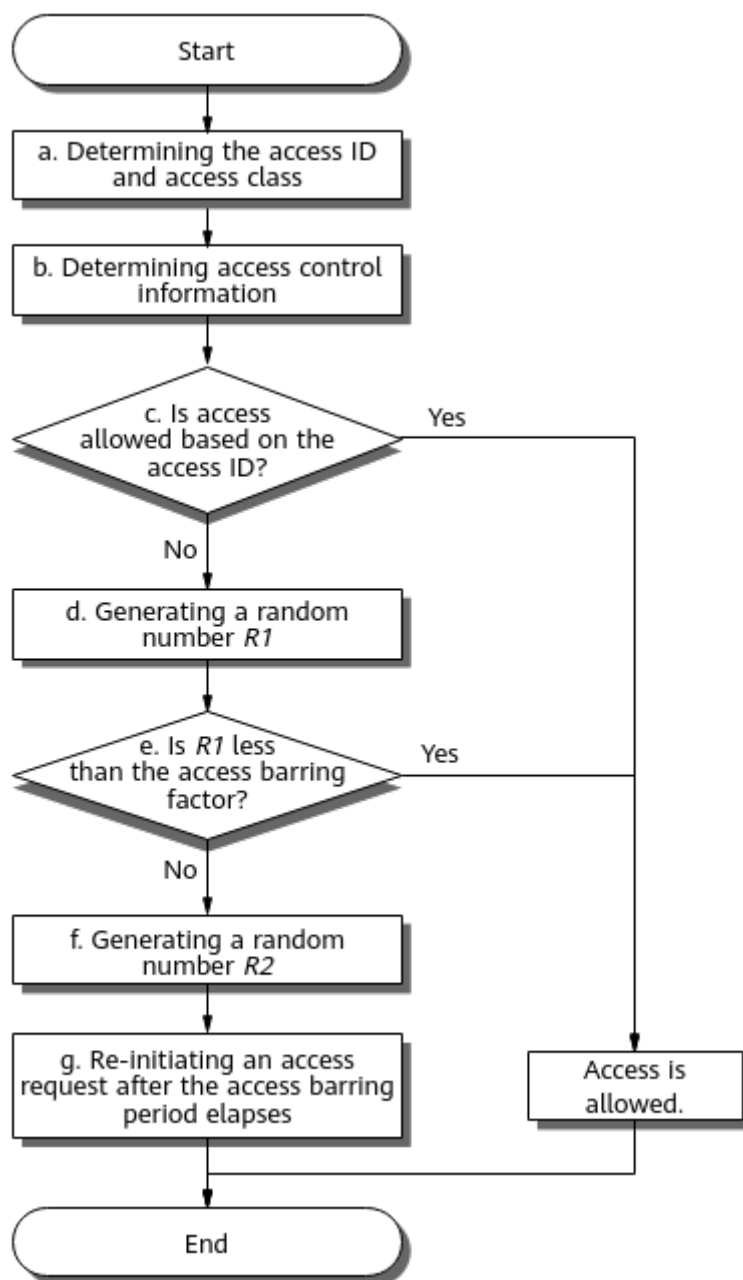
4.1 Principles

4.1.1 UAC Process

The gNodeB delivers access control parameters through SIB1 to perform UAC on UEs in idle or inactive mode in a cell. UAC is implemented by the gNodeB together with UEs. The basic process is as follows:

1. The gNodeB broadcasts access control information in SIB1 using any of the following methods:
 - Static access class control: When the **NRDUCellUacBar.AccessControlType** parameter is set to **STATIC_UAC**, static access class control takes effect. Access control-related parameters configured by running MML commands are broadcast to UEs through SIB1. For details, see [4.1.2 Static Access Class Control](#).
 - Dynamic access class control: When the **NRDUCellUacBar.AccessControlType** parameter is set to **DYNAMIC_UAC**, the gNodeB dynamically controls whether to broadcast access control information in SIB1 to UEs based on the system load. For details, see [4.1.3 Dynamic Access Class Control](#).
2. Before initiating an RRC connection setup or RRC connection resume procedure, a UE determines whether it can access the cell based on the access control information broadcast in SIB1, as shown in [Figure 4-1](#).

Figure 4-1 Procedure for a UE to determine whether access is allowed during access class control



The procedure is as follows:

- a. Determining the access ID and access class: The access IDs and access classes corresponding to the access are determined.
- b. Determining access control information: Based on the operator and access service type, the UE finds the access control parameters specific to the access from the *uac-BarringInfo* IE in SIB1.
- c. Determine whether access is allowed based on the access ID: The UE is allowed to access the network if the access barring status corresponding to at least one of the UE's access IDs is allowed. Otherwise, the UE proceeds to **d**.

- d. Generating a random number *R1*: The UE generates a random number *R1* ($0 \leq R1 < 1$).
- e. Checking whether *R1* is less than the access barring factor: If *R1* is less than the access barring factor, the UE is allowed to access the network. Otherwise, the UE proceeds to **f**. The **gNBUEacParamGroup.BarringFactor** parameter specifies the access barring factor, and its value ranges from 0% to 95%.
- f. Generating a random number *R2*: The UE generates a random number *R2* ($0 \leq R2 < 1$).
- g. After the access barring period elapses, the UE re-initiates an access request. The access barring period is calculated by using the following formula: $(0.7 + 0.6 \times R2) \times uac-BarringTime$. *uac-BarringTime* indicates the access barring period, which ranges from 4s to 512s, and is configured by using the **gNBUEacParamGroup.BarringTime** parameter.

For more information about the procedure for a UE to determine whether access is allowed, see section 5.3.14 "Unified Access Control" in 3GPP TS 38.331 V15.6.0.

4.1.2 Static Access Class Control

When the **NRDUCellUacBar.AccessControlType** parameter is set to **STATIC_UAC**, static access class control takes effect. Static access class control is used to mitigate traffic congestion in special scenarios such as festivals, earthquakes, and fires. It consists of two steps:

- 1. Adding a base-station-level access control parameter group
The **gNBUEacParamGroup** MO is used to configure an access control parameter group, including parameters such as **Barring State for Special Access ID**, **Barring Factor**, and **Barring Time**. **Table 4-1** lists these parameters.

Table 4-1 Access control parameters in the gNBUEacParamGroup MO

Parameter Name	Parameter ID	Parameter Description
Access Control Parameter Group ID	gNBUEacParamGroup.AccessControlParamGrpId	Indicates the ID of an access control parameter group.

Parameter Name	Parameter ID	Parameter Description
Barring State for Special Access ID	gNB UacParamGroup. BarStateForSpecialAcsId	Indicates whether access barring check is required for UEs with a special access ID^a. For details about the access IDs of UEs, see section 6.22.2.2 "Access identities" in 3GPP TS 22.261 V15.6.0. ACCESS_ID1_BAR_STATE: If this option is selected, access barring check is performed for UEs with access ID 1, and whether the access attempts are allowed is determined based on the check result. If this option is deselected, the access attempts of UEs with access ID 1 are allowed. ACCESS_ID2_BAR_STATE: If this option is selected, access barring check is performed for UEs with access ID 2, and whether the access attempts are allowed is determined based on the check result. If this option is deselected, the access attempts of UEs with access ID 2 are allowed. ACCESS_ID11_BAR_STATE: If this option is selected, access barring check is performed for UEs with access ID 11, and whether the access attempts are allowed is determined based on the check result. If this option is deselected, the access attempts of UEs with access ID 11 are allowed. ACCESS_ID12_BAR_STATE: If this option is selected, access barring check is performed for UEs with access ID 12, and whether the access attempts are allowed is determined based on the check result. If this option is deselected, the access attempts of UEs with access ID 12 are allowed. ACCESS_ID13_BAR_STATE: If this option is selected, access barring check is performed for UEs with access ID 13, and whether the access attempts are allowed is determined based on the check result. If this

Parameter Name	Parameter ID	Parameter Description
		option is deselected, the access attempts of UEs with access ID 13 are allowed. ACCESS_ID14_BAR_STATE: If this option is selected, access barring check is performed for UEs with access ID 14, and whether the access attempts are allowed is determined based on the check result. If this option is deselected, the access attempts of UEs with access ID 14 are allowed. ACCESS_ID15_BAR_STATE: If this option is selected, access barring check is performed for UEs with access ID 15, and whether the access attempts are allowed is determined based on the check result. If this option is deselected, the access attempts of UEs with access ID 15 are allowed.
Barring Factor	gNBuacParamGroup.BarringFactor	Indicates the access barring factor for an access category. A random number ($R1$) is generated by a UE after it initiates an access request. If the random number is less than the access barring factor, UE access is allowed. If the random number is greater than or equal to the access barring factor, UE access is barred. The value of this parameter ranges from 0% to 95%. For details, see section 5.3.14.5 "Access barring check" in 3GPP TS 38.331 V15.10.0.
Barring Time	gNBuacParamGroup.BarringTime	Indicates the access barring duration (indicated by <i>uac-BarringTime</i>) for an access category. The access barring duration of a UE is calculated using the following formula: $(0.7 + 0.6 \times R2) \times uac-BarringTime$. After the access barring duration elapses, the UE makes the access attempt again. The value of this parameter ranges from 4s to 512s. For details, see section 5.3.14.5 "Access barring check" in 3GPP TS 38.331 V15.10.0.

Parameter Name	Parameter ID	Parameter Description
a: Provided that the other configuration steps described in this document are completed, access barring check is always required for common UEs (with access ID 0), regardless of the gNB UacParamGroup. <i>BarStateForSpecialAcsl</i> d parameter setting.		

2. Adding a mapping between an access class and an access control parameter group for a cell

The **NRDUCellUacBar** MO is used to configure cell access control information. **Table 4-2** lists the related parameters in this MO. The access control type, access control entering and exiting thresholds, and referenced access control parameter group can be specifically configured for each cell based on operators and access classes (service types). Static access class control requires that the **NRDUCellUacBar**.*AccessControlType* parameter be set to **STATIC_UAC**.

Table 4-2 Access control parameters in the NRDUCellUacBar MO

Parameter Name	Parameter ID	Parameter Description
Operator ID	NRDUCellUacBar . <i>OperatorId</i>	Indicates the ID of an operator. If this parameter is set to a value in the range from 0 to 5, the access control policy is configured for a specific operator. If this parameter is set to 255 , the access control policy applies to operators whose policies are not configured.
Access Category	NRDUCellUacBar . <i>AccessCategory</i>	Indicates the access class that requires access control in a cell. For details, see section 6.22.2.3 "Access categories" in 3GPP TS 22.261 V15.6.0. According to section 5.3.14 "Unified Access Control" in 3GPP TS 38.331 V15.6.0, access class control is not required for UEs of access class 0. Therefore, only the access control information of access classes 1 to 63 needs to be delivered.

Parameter Name	Parameter ID	Parameter Description
Access Control Type	NRDUCellUacBar.AccessControlType	Indicates the access control type. If this parameter is set to STATIC_UAC , the gNodeB broadcasts configured access control parameters in SIB1. If this parameter is set to DYNAMIC_UAC , the gNodeB adjusts the access control parameters broadcast in SIB1 in real time based on the system load.
Access Control Parameter Group ID	NRDUCellUacBar.AccessControlParamGrpId	Indicates the access control parameter group ID mapping to an access class.

4.1.3 Dynamic Access Class Control

When the **NRDUCellUacBar.AccessControlType** parameter is set to **DYNAMIC_UAC**, dynamic access class control takes effect. The gNodeB periodically evaluates whether to trigger access class control based on the current flow control state or the number of RRC_CONNECTED UEs. If the **BARRING_FACTOR_ADAPT_SW** option of the **NRDUCellAlgoSwitch.RacAlgoSwitch** parameter is selected, the evaluation period is 20s. Otherwise, the evaluation period is 60s.

The gNodeB determines that a cell enters the congestion state when at least one of the following conditions is met upon expiration of a period:

- In the period, the proportion of the flow control duration is greater than the threshold for entering UAC based on flow control. The threshold is specified by the **NRDUCellUacBar.FlowCtrlUacEnterThld** parameter.
- In the period, the proportion of the number of RRC_CONNECTED UEs to the maximum number of UEs in a cell is greater than the threshold for entering UAC based on the number of RRC_CONNECTED UEs. The threshold is specified by the **NRDUCellUacBar.RrcUserNumUacEnterThld** parameter.

The gNodeB determines that a cell recovers from congestion when all of the following conditions are met upon expiration of a period:

- In the period, the proportion of the flow control duration is less than the threshold for exiting UAC based on flow control. The threshold is specified by the **NRDUCellUacBar.FlowCtrlUacExitThld** parameter.
- In the period, the proportion of the number of RRC_CONNECTED UEs to the maximum number of UEs in a cell is less than the threshold for exiting UAC based on the number of RRC_CONNECTED UEs. The threshold is specified by the **NRDUCellUacBar.RrcUserNumUacExitThld** parameter.

Upon expiration of a period, the gNodeB determines that a cell's state (congested or not) retains the same as that in the last period when at least one of the following conditions is met:

- In the period, the proportion of the flow control duration is greater than the threshold for exiting UAC based on flow control (specified by the **NRDUCellUacBar.FlowCtrlUacExitThld** parameter) but less than the threshold for entering UAC based on flow control (specified by the **NRDUCellUacBar.FlowCtrlUacEnterThld** parameter).
- In the period, the proportion of the number of RRC_CONNECTED UEs to the maximum number of UEs in a cell is greater than the threshold for exiting UAC based on the number of RRC_CONNECTED UEs (specified by the **NRDUCellUacBar.RrcUserNumUacExitThld** parameter) but less than the threshold for entering UAC based on the number of RRC_CONNECTED UEs (specified by the **NRDUCellUacBar.RrcUserNumUacEnterThld** parameter).

In the preceding conditions:

- The **NRDUCellUacBar.RrcUserNumCountMode** parameter specifies the measurement mode of the number of RRC_CONNECTED UEs.
 - If this parameter is set to **PLMN_MODE**, the number of RRC_CONNECTED UEs is calculated on a per operator basis in a cell.
 - If this parameter is set to **CELL_MODE**, the number of RRC_CONNECTED UEs is calculated on a per cell basis, including the number of RRC_CONNECTED UEs of all operators in the cell.

The **NRDUCellUacBar.RrcUserNumCountMode** parameter takes effect only when UAC based on the number of RRC_CONNECTED UEs is in effect.

- The proportion of flow control duration and the maximum number of UEs in a cell are calculated on a per cell basis. The "maximum number of UEs in a cell" refers to the maximum number of UEs in RRC connected mode in a cell. For details, see the UE number specifications of the corresponding RAT in "NR Technical Specifications of Baseband Processing Units" in "BBU Technical Specifications" in *3900 & 5900 Series Base Station Product Documentation*.
- Threshold parameters for dynamic access class control can be configured on a per cell, per operator, or per access class basis. For details, see [Table 4-3](#).

Table 4-3 Threshold parameters for dynamic access class control

Parameter Name	Parameter ID	Parameter Description
Flow Control UAC Enter Threshold	NRDUCellUacBar.FlowCtrlUacEnterThld	Indicates the threshold for entering UAC based on flow control. If the proportion of the flow control duration is greater than this threshold, the gNodeB broadcasts the access control parameters of the corresponding access class in SIB1. When this parameter is set to 255 , UAC based on flow control is not enabled.

Parameter Name	Parameter ID	Parameter Description
Flow Control UAC Exit Threshold	NRDUCellUacBar.FlowCtrlUacExitThld	Indicates the threshold for exiting UAC based on flow control. If the proportion of the flow control duration is less than this threshold, the gNodeB does not broadcast access control parameters of the corresponding access class in SIB1. When this parameter is set to 255 , UAC based on flow control is not enabled.
RRC User Number UAC Enter Threshold	NRDUCellUacBar.RrcUserNumUacEnterThld	Indicates the threshold for entering UAC based on the number of RRC_CONNECTED UEs. If the proportion of the number of RRC_CONNECTED UEs to the maximum number of UEs in a cell is greater than this threshold, the gNodeB broadcasts access control parameters of the corresponding access class in SIB1. When this parameter is set to 255 , UAC based on the number of RRC_CONNECTED UEs is not enabled.
RRC User Number UAC Exit Threshold	NRDUCellUacBar.RrcUserNumUacExitThld	Indicates the threshold for exiting UAC based on the number of RRC_CONNECTED UEs. If the proportion of the number of RRC_CONNECTED UEs to the maximum number of UEs in a cell is less than this threshold, the gNodeB does not broadcast access control parameters of the corresponding access class in SIB1. When this parameter is set to 255 , UAC based on the number of RRC_CONNECTED UEs is not enabled.
Note: Set the parameters in this table when the NRDUCellUacBar.AccessControlType parameter is set to DYNAMIC_UAC .		

When the **NRDUCellUacBar.AccessControlType** parameter is set to **DYNAMIC_UAC**, the parameters in at least one of the following two parameter groups must be set to valid values ranging from 0 to 100:

- Parameter group 1: includes the threshold for entering UAC based on flow control (**NRDUCellUacBar.FlowCtrlUacEnterThld**) and the threshold for exiting UAC based on flow control (**NRDUCellUacBar.FlowCtrlUacExitThld**).

- Parameter group 2: includes the threshold for entering UAC based on the number of RRC_CONNECTED UEs (**NRDUCellUacBar.RrcUserNumUacEnterThld**) and the threshold for exiting UAC based on the number of RRC_CONNECTED UEs (**NRDUCellUacBar.RrcUserNumUacExitThld**).

 NOTE

In each parameter group, the entering and exiting thresholds must meet any of the following conditions. Otherwise, the parameter group does not take effect.

- Both the entering and exiting thresholds are set to valid values ranging from 0 to 100, and the entering threshold must be greater than the exiting threshold.
- Both the entering and exiting thresholds are set to the invalid value **255**.

Access Barring Factor Adaptation

To optimize access control effects for congested cells in FDD or high-frequency TDD scenarios, the access barring factor adaptation function is introduced. This function is controlled by the **BARRING_FACTOR_ADAPT_SW** option of the **NRDUCellAlgoSwitch.RacAlgoSwitch** parameter. When this option is selected, this function takes effect. After this function takes effect, the gNodeB adaptively adjusts the access barring factor based on the current flow control state or the number of RRC_CONNECTED UEs. The adjustment rules are as follows:

- When a cell meets the conditions for entering the congestion state, the access barring factor is adjusted to the minimum value, which is specified by the **gNBuacParamGroup.MinBarringFactor** parameter.
- When a cell meets the conditions for recovering from congestion, the access barring factor is increased by one step. If the current access barring factor has reached the maximum value P95 (95%), the cell is considered to be recovered from congestion.
- Otherwise, the access barring factor retains the same as that in the last evaluation period.

 NOTE

When the access barring factor adaptation function is enabled, the following rules apply:

- If the **NRDUCellUacBar.AccessControlParamGrpId** parameter (specifying an access control parameter group ID) is set to the same value in different **NRDUCellUacBar** MO instances for a cell, the following parameters must be set to the same values: **NRDUCellUacBar.RrcUserNumCountMode**, **NRDUCellUacBar.FlowCtrlUacEnterThld**, **NRDUCellUacBar.RrcUserNumUacEnterThld**, **NRDUCellUacBar.FlowCtrlUacExitThld**, and **NRDUCellUacBar.RrcUserNumUacExitThld**.
- If the **NRDUCellUacBar.OperatorId** parameter (specifying an operator ID) is set to different values in different **NRDUCellUacBar** MO instances for a cell, the corresponding **NRDUCellUacBar.AccessControlParamGrpId** parameter (specifying an access control parameter group ID) must be set to different values.

4.2 Network Analysis

4.2.1 Benefits

When a large number of UEs simultaneously access the network, this function performs control on cell access of UEs in idle or inactive mode, thereby reducing network congestion.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- Static access class control

On uncontested networks, this function reduces the traffic volume. On congested networks, this function relieves cell congestion, thereby increasing the success rate of RRC connection setup.

- Dynamic access class control

When the proportion of the flow control duration or the proportion of RRC_CONNECTED UEs is greater than a specified threshold, this function gradually blocks some UEs from accessing the cell, thereby relieving cell congestion and increasing the success rate of RRC connection setup.

When a cell is severely congested, the access barring factor adaptation function increases the RRC connection setup success rate. In other scenarios, this function has no impact.

NOTE

For details about flow control, see *Flow Control*.

In multi-operator scenarios, if the **NRDUCellUacBar.RrcUserNumCountMode** parameter is set to **CELL_MODE**, UAC based on the number of RRC_CONNECTED UEs is more likely to be triggered, leading to a more significant effect of relieving system congestion but a greater impact on user experience; if the **NRDUCellUacBar.RrcUserNumCountMode** parameter is set to **PLMN_MODE**, UAC based on the number of RRC_CONNECTED UEs is less likely to be triggered, leading to a less significant effect of relieving system congestion but a smaller impact on user experience. In single-operator scenarios, the values **CELL_MODE** and **PLMN_MODE** provide the same effects.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Dynamic access class control	NRDUCellUacBar.AccessControlType	<i>Access Class Control</i>	Enabling access barring factor adaptation requires that dynamic access class control take effect (NRDUCellUacBar.AccessControlType set to DYNAMIC_UAC).

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Operator ID	NRDUCellUacBar.OperatorId	<i>Access Class Control</i>	When the NRDUCellUacBar.OperatorId parameter is set to 255, the NRDUCellUacBar.RrcUserNumCountMode parameter must be set to CELL_MODE .

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

[Table 4-4](#) and [Table 4-5](#) describe the parameters used for function activation and optimization, respectively.

Table 4-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellUacBar.NrDuCellId	Set this parameter based on the network plan.
Operator ID	NRDUCellUacBar.OperatorId	Set this parameter based on the network plan.
Access Category	NRDUCellUacBar.AccessCategory	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Access Control Type	NRDUCellUacBar.AccessControlType	Set this parameter based on the network plan.
Access Control Parameter Group ID	NRDUCellUacBar.AccessControlParamGrpId	Set this parameter based on the network plan.
Access Control Parameter Group ID	gNBuacParamGroup.AccessControlParamGrpId	Set this parameter based on the network plan.
RRC User Number Count Mode	NRDUCellUacBar.RrcUserNumCountMode	Set this parameter based on the network plan.

Table 4-5 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Flow Control UAC Enter Threshold	NRDUCellUacBar.FlowCtrlUacEnterThld	None	This parameter takes effect only when NRDUCellUacBar.AccessControlType is set to DYNAMIC_UAC . Set this parameter based on the network plan.
Flow Control UAC Exit Threshold	NRDUCellUacBar.FlowCtrlUacExitThld	None	This parameter takes effect only when NRDUCellUacBar.AccessControlType is set to DYNAMIC_UAC . Set this parameter based on the network plan.
RRC User Number UAC Enter Threshold	NRDUCellUacBar.RrcUserNumUacEnterThld	None	This parameter takes effect only when NRDUCellUacBar.AccessControlType is set to DYNAMIC_UAC . Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
RRC User Number UAC Exit Threshold	NRDUCellUacBar.RrcUserNumUacExitThld	None	This parameter takes effect only when NRDUCellUacBar.AccessControlType is set to DYNAMIC_UAC . Set this parameter based on the network plan.
Barring State for Special Access ID	gNBuacParamGroup.BarStateForSpecialAcId	None	Set this parameter based on the network plan.
Barring Factor	gNBuacParamGroup.BarringFactor	None	Set this parameter based on the network plan.
Barring Time	gNBuacParamGroup.BarringTime	None	Set this parameter based on the network plan.
RAC Algorithm Switch	NRDUCellAlgoSwitch.RacAlgoSwitch	BARRING_FACT OR_ADAPT_SW	Set this parameter based on the network plan.
Minimum Barring Factor	gNBuacParamGroup.MinBarringFactor	None	Set this parameter based on the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Adding an access control parameter group
ADD GNBuacPARAMGROUP: AccessControlParamGrpId=0,
BarStateForSpecialAcId=ACCESS_ID1_BAR_STATE-1, BarringFactor=P50, BarringTime=S4;
ADD GNBuacPARAMGROUP: AccessControlParamGrpId=1,
BarStateForSpecialAcId=ACCESS_ID2_BAR_STATE-1, BarringFactor=P60, BarringTime=S8;
//Setting access control parameters for UEs of access category 1 for all operators in the cell and setting
AccessControlType to STATIC_UAC
ADD NRDUCellUACBAR: NrDuCellId=0, OperatorId=255, AccessCategory=1, AccessControlParamGrpId=0,
```

```
AccessControlType=STATIC_UAC;
//Setting access control parameters for UEs of access category 2 for all operators in the cell and setting
AccessControlType to DYNAMIC_UAC
ADD NRDUCELLUACBAR: NrDuCellId=0, OperatorId=255, AccessCategory=2, AccessControlParamGrpId=1,
AccessControlType=DYNAMIC_UAC, FlowCtrlUacEnterThld=70, FlowCtrlUacExitThld=50,
RrcUserNumUacEnterThld=70, RrcUserNumUacExitThld=50, RrcUserNumCountMode=CELL_MODE;
```

Optimization Command Examples

```
//Modifying the access control parameter group
MOD GNBACPARAMGROUP: AccessControlParamGrpId=0, BarringFactor=P70, BarringTime=S16;
//Modifying parameter configurations for static access control
MOD NRDUCELLUACBAR: NrDuCellId=0, OperatorId=255, AccessCategory=2, AccessControlParamGrpId=1,
AccessControlType=STATIC_UAC;
//Modifying parameter configurations for dynamic access control
MOD NRDUCELLUACBAR: NrDuCellId=0, OperatorId=255, AccessCategory=1, AccessControlParamGrpId=0,
AccessControlType=DYNAMIC_UAC, FlowCtrlUacEnterThld=255, FlowCtrlUacExitThld=255,
RrcUserNumUacEnterThld=70, RrcUserNumUacExitThld=50;
MOD NRDUCELLUACBAR: NrDuCellId=0, OperatorId=255, AccessCategory=2, AccessControlParamGrpId=1,
AccessControlType=DYNAMIC_UAC, FlowCtrlUacEnterThld=255, FlowCtrlUacExitThld=255,
RrcUserNumUacEnterThld=70, RrcUserNumUacExitThld=50;
//Turning on the switch for access barring factor adaptation
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RacAlgoSwitch=BARRING_FACTOR_ADAPT_SW-1;
//Modifying the minimum access barring factor
MOD GNBACPARAMGROUP: AccessControlParamGrpId=1, MinBarringFactor=P30;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Removing access control parameter settings
RMV NRDUCELLUACBAR: NrDuCellId=0, OperatorId=255, AccessCategory=1;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Method 1

Step 1 Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.

Step 2 Check whether the *uac-BarringInfo* IE is included in SIB1. If included, the unified access control function has taken effect.

----End

Method 2

Run the **DSP NRDUCELLUACBAR** command to query the effective state. If the value of **Flow Control-based UAC State** or **RRC_CONNECTED UE Number-based UAC State** in **Access Control State** is **On** in the command output, the dynamic

access control function has taken effect. If the value of **STATIC_UAC** in **Access Control State** is **On** in the command output, the static access control function has taken effect.

Method 3

Observe performance counters to check whether dynamic access control has taken effect:

- If the value of the **N.DynACBar.Dur** counter is not 0, the dynamic access control function has taken effect in the cell.
- If the value of the **N.DynACBar.Dur.PLMN** counter is not 0, the dynamic access control function has taken effect for the operator of the specified PLMN in the cell.

4.4.3 Network Monitoring

After the access control function takes effect, the values returned by the counters listed in **Table 4-6** will decrease.

Table 4-6 Counters related to access control

Counter ID	Counter Name
1911816954	N.RRC.SetupReq.Att
1911816955	N.RRC.SetupReq.Att.MoVoiceCall
1911816957	N.RRC.SetupReq.Succ.HighPri
1911817825	N.RRC.SetupReq.Att.MoVideoCall
1911817826	N.RRC.SetupReq.Att.MoSms
1911817827	N.RRC.SetupReq.Att.MpsPri
1911817832	N.RRC.SetupReq.Att.McsPri
1911817001	N.RRC.SetupReq.Msg.Disc.FlowCtrl

5 Access Control Based on Operator-specific Reservation

5.1 Principles

When a cell is reserved for an operator, the gNodeB broadcasts the cell reservation information in SIB1. During cell selection or reselection, UEs of access class 11 or 15 in their HPLMN/EHPLMN treat this cell as a candidate cell, while other UEs treat this cell as a barred cell. This significantly reduces the load of the cell.

When **NRDUCellSelConfig.CellReservedForOp** is set to **RESERVED**, or when **NRDUCellSelConfig.CellReservedForOp** is set to **NOT_RESERVED** and **NRDUCellOp.CellReservedForOp** is set to **RESERVED**, the cell is reserved for an operator.

- The UEs of access class 11 or 15 in their HPLMN or EHPLMN treat this cell as a candidate cell during cell selection or reselection.
- The UEs of access classes 0 to 2 or access classes 12 to 14, or the UEs of access class 11 or 15 outside their HPLMN/EHPLMN treat this cell as a barred cell during cell selection or reselection.

When **NRDUCellSelConfig.CellReservedForOp** is set to **NOT_RESERVED**, and **NRDUCellOp.CellReservedForOp** is set to **NOT_RESERVED** or **NRDUCellOp.CellReservedForOp** is left unspecified, the cell is not reserved for operator use. In this case, all UEs treat this cell as a candidate cell during cell selection or reselection.

5.2 Network Analysis

5.2.1 Benefits

In the case of network congestion, this function can relieve cell congestion and significantly reduce the load of the local cell.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function may reduce the new traffic volume on the network. If a cell is reserved for an operator, UEs, except those of access class 11 or 15 in their HPLMN/EHPLMN, treat this cell as a barred cell during cell selection or reselection.

5.3 Requirements

5.3.1 Licenses

There are no license requirements for basic functions.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-1 describes the parameters used for function activation.

Table 5-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR DU Cell ID	NRDUCellSelConfig.NrDuCellId	Set this parameter based on the network plan.
Cell Reserved for Operator	NRDUCellSelConfig.CellReservedForOp	Set this parameter based on the network plan.
NR DU Cell ID	NRDUCellOp.NrDuCellId	Set this parameter based on the network plan.
Operator ID	NRDUCellOp.OperatorId	Set this parameter based on the network plan.
Cell Reserved for Operator	NRDUCellOp.CellReservedForOp	Set this parameter based on the network plan.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring a cell to be reserved for all operators  
MOD NRDUCELLSELCONFIG: NrDuCellId=0, CellReservedForOp=RESERVED;  
//Configuring a cell to be reserved for a specified operator  
MOD NRDUCELLSELCONFIG: NrDuCellId=0, CellReservedForOp=NOT_RESERVED;  
MOD NRDUCELLLOP: NrDuCellId=0, OperatorId=1, CellReservedForOp=RESERVED;
```

Optimization Command Examples

None

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Configuring a cell to be not reserved for a specified operator  
MOD NRDUCELLSELCONFIG: NrDuCellId=0, CellReservedForOp=NOT_RESERVED;  
MOD NRDUCELLLOP: NrDuCellId=0, OperatorId=1, CellReservedForOp=NOT_RESERVED;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

- Step 1** Start Uu signaling tracing: Log in to the MAE-Access and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
- Step 2** Check whether the value of the *cellReservedForOperatorUse* IE in SIB1 is **reserved**. If the value is **reserved**, this function has taken effect.

----End

5.4.3 Network Monitoring

After this function takes effect, the values of the counters listed in [Table 5-2](#) will decrease.

Table 5-2 Counters related to access control

Counter ID	Counter Name
1911816954	N.RRC.SetupReq.Att
1911816955	N.RRC.SetupReq.Att.MoVoiceCall
1911816957	N.RRC.SetupReq.Succ.HighPri
1911817825	N.RRC.SetupReq.Att.MoVideoCall
1911817826	N.RRC.SetupReq.Att.MoSms
1911817827	N.RRC.SetupReq.Att.MpsPri
1911817832	N.RRC.SetupReq.Att.McsPri
1911817001	N.RRC.SetupReq.Msg.Disc.FlowCtrl

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 22.261: "Service requirements for the 5G system; Stage 1"
- 3GPP TS 24.501: "Non-Access-Stratum (NAS) protocol for 5G System (5GS); Stage 3"
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*